Lecture 08, Math 441

Aldo Manfroi

University of Illinois at Urbana-Champaign

Population Models and Autonomous Equations

Assume the rate of change of $P(t)$, a population of fish in a pond, is given by

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \alpha P - \beta P = \gamma P,$$

where the αP represents an increase due to births, and $-\beta P$ is a decrease due to deaths by old age, and $\gamma = \alpha - \beta$. The solution is just $P(t) = P_0 \mathrm{e}^{\gamma t}$, where P_0 is the population at time 0.

Logistic Equation

γ is a constant growth rate. Let's now make it a function of P itself, for example: $\gamma(P) = R(P^* - P)$, where R is a constant.

The population equation is now

$$\frac{\mathrm{d}P}{\mathrm{d}t} = R(P^* - P)\,P = RP^*P - RP^2 = \mathcal{F}(P).$$

This is an example of an autonomous equation, an equation that does not explicitly depend on the independent variable.

timestamp: 03:55

An equivalent form is

$$\frac{\mathrm{d}P}{\mathrm{d}t} = R\left(1 - \frac{P}{K}\right) P.$$

The growth rate is positive if $P > K$ and negative if $P < K$.

This is an example of a logistic equation. The more general form is

$$\frac{\mathrm{d}P}{\mathrm{d}t} = \sigma P - \delta P^2, \text{ where } \{\sigma, \delta\} > 0$$

timestamp: 07:35

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = 2P - P^2 = (2-P)\,P$. This can be separated into:

$$\frac{\mathrm{d}P}{(2-P)P} = \mathrm{d}t,$$

assuming that $P \neq 0, 2$. To integrate the left hand side we use partial fractions:

$$\frac{1}{(2-P)P} = \frac{A}{P} + \frac{B}{2-P} = \frac{A(2-P)+BP}{(2-P)P} = \frac{2A+(B-A)P}{(2-P)P}$$

This works if $A = B = \frac{1}{2}$

The equation can be rewritten and solved as

$$\left(\frac{1}{2P}+\frac{1}{2(2-P)}\right)\mathrm{d}P=\mathrm{d}t \;\rightarrow\; \left(\frac{1}{P}+\frac{1}{2-P}\right)\mathrm{d}P=2\mathrm{d}t \;\xrightarrow{\int}$$

$$\ln(P)-\ln(2-P)=2t+C \;\rightarrow\; \ln\left(\frac{P}{2-P}\right)=2t+C$$

and after taking the exponential of both sides and with some algebra:

$$P(t)=\frac{2D\,\mathrm{e}^{2t}}{1+D\,\mathrm{e}^{2t}}=\frac{2D}{D+\mathrm{e}^{-2t}},\ \text{where } P\neq\{0,2\}$$

timestamp: 12:33

Singular Solutions

This solution was obtained with the assumption $P \neq \{0, 2\}$. But the functions $P(t) = 0$ and $P(t) = 2$ actually satisfy the equation. These are singular solutions, not included in the general solution found above.

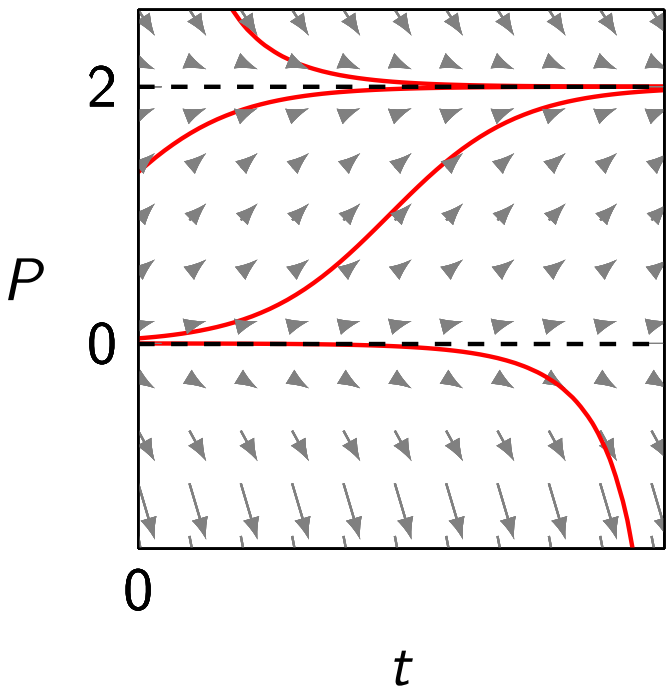
Slope Fields

To study the behavior of the solutions it is helpful to go back to the original equation $\frac{\mathrm{d}P}{\mathrm{d}t} = 2P - P^2 = (2 - P)\,P$

The right hand side gives the slope of the solutions, since it is equal to the rate of change, so the slope field is obtained by calculating the value of the r.h.s. for different values of P.

timestamp: 24:23

$$\frac{dP}{dt} = (2 - P)\,P$$



For $P = 0$ and $P = 2$ the slope is 0.

Such values of P are called

critical points and correspond to

equilibrium solutions because they

don't change with time.

The other solutions curves are in

between, above, and below these

equilibriums.

timestamp: 25:00

The slope field or direction field gives us a graphical representation of what the solutions of the equation do. We can use the slope field to draw different solution curves.

In this example $P(t) = 2$ is a stable equilibrium solution because solutions near it tend to go toward it as time grows.

$P(t) = 0$ is an unstable equilibrium solution because solutions near it tend to go away from it as time grows.

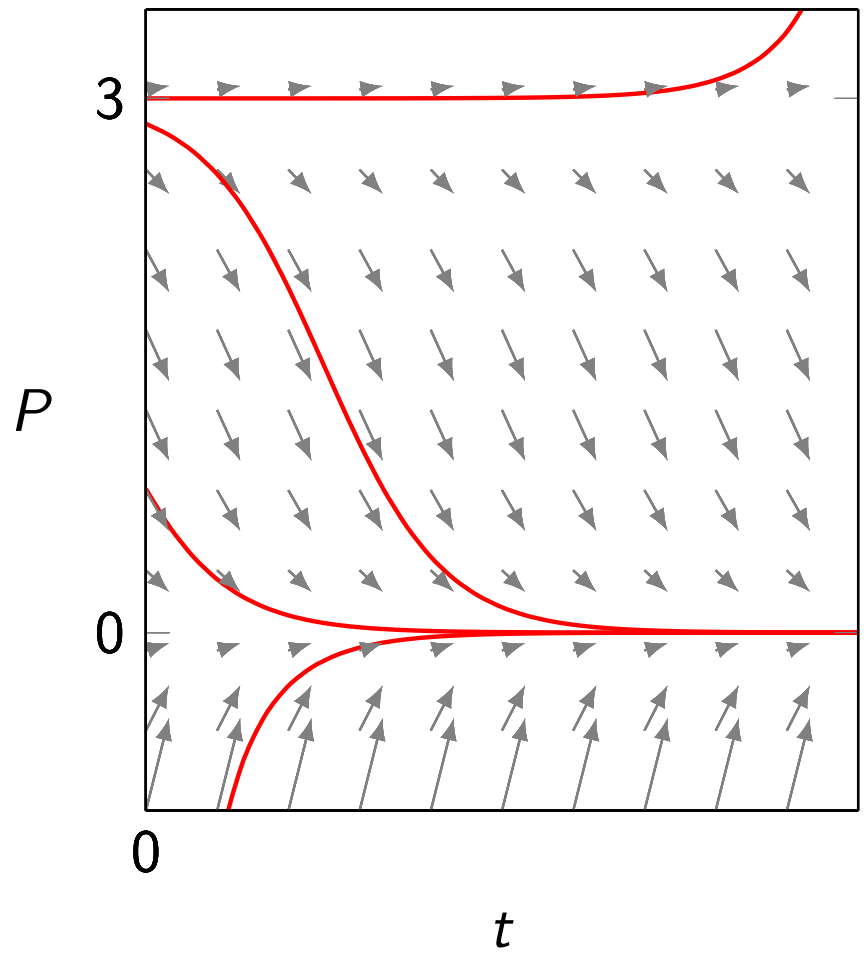
timestamp: 29:00

Note: none of the solution curves can cross or touch other solution curves, because that would violate the theorem of existence and uniqueness which applies to this equation.

timestamp: 33:40

Ex: $\frac{\mathrm{d}P}{\mathrm{d}t} = P^2 - 3P = P(P-3)$

$$\frac{\mathrm{d}P}{\mathrm{d}t} = P(P-3)$$



timestamp: 36:20

Phase Diagram

We can summarize the behavior of the solutions with a single axis with a phase diagram.

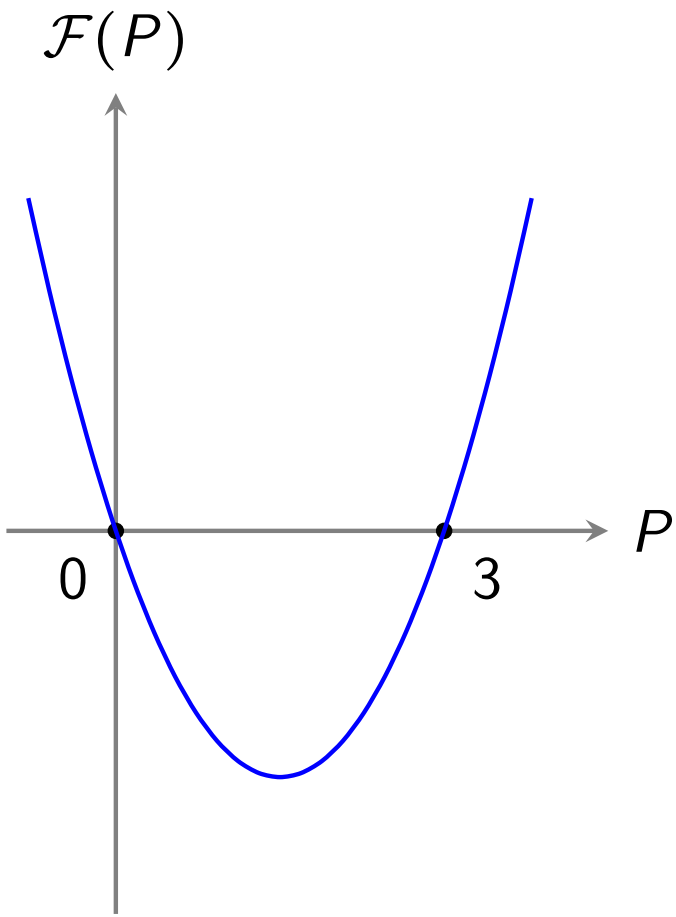


$$\frac{dP}{dt} = P(P-3)$$

Where the arrows indicates the direction of the solution curve's slope in each region.

timestamp: 40:50

Another useful graph is obtained by plotting the value of the right hand side (in this example $P^2 - 3P$) as a function of P.



Where the sign of $\mathcal{F}(P)$ shows the solution curve's slope in each region.

timestamp: 42:06